

SCIENTIFIC REFEREE REPORT

The Face-Adjacency Graph of the Truncated Octahedron

Laplacian spectrum, symmetry decomposition, an inter-orbit operator, and the magnetic Laplacian

Luke Martin — **Draft 9, October 2026**

Assessment scope: 14-page Draft 9 PDF; targeted comparison against Draft 8; inspection of the public pinned Lean proof files and previous bibliographic sources; independent exact checks of the graph, magnetic and orbit-basis matrices, and the topological premise. Neither the complete tagged-release runner nor the Lean kernel was executed in this reviewing environment.

Recommendation: **Minor revision — conditional on independent release replay.** Draft 9 resolves the previous mathematical-exposition and prior-art requests and provides credible, narrowly scoped formal proof artifacts. Final computational certification is withheld pending a clean tagged Python and Lean run.

Mathematical validity gate: **No confirmed critical or major error in the examined claims.** The review does not certify every theorem, every formal statement, or the unexecuted release.

Outstanding matters: One release-level closure requirement (including the Lean build); one important formal-proof scope boundary, accurately disclosed by the author, to preserve in the final manuscript.

Aggregate score: Omitted. The substantive findings and fourteen individually justified criteria determine this recommendation.

Executive assessment

Draft 9 is a focused mathematical note giving exact Laplacian and special-flux magnetic spectra for the face-adjacency graph of the truncated octahedron. Its chief substantive change is not a new numerical spectrum: it introduces Lean 4 formal proofs for the explicit ordinary Laplacian and for the $q = 6$ and $q = 12$ magnetic characteristic polynomials, together with orbit-basis identities for the antisymmetric operator K . The authors have also incorporated a precise chain-complex justification for Proposition 6.1 and an explicit negative control illustrating the necessity of the spherical hypothesis.

The new formal source was inspected in the tagged public repository: in particular, `Theorem31.lean` gives an explicit rational 14×14 Laplacian and a conjugation-to-blocks argument; `HalfFlux.lean` and `QuarterFlux.lean` state the two complete characteristic-polynomial identities; and `Theorem51.lean` proves the stated rational-basis relations for K . The source and its README claim kernel checking with only standard Lean axioms. This review did *not* independently rerun that kernel, and consequently assesses these as documented formal artifacts whose reported build remains to be reproduced.

Independent exact reconstruction of the same graph, two magnetic matrices, K and the oriented boundary maps passed 25 targeted checks. The tests included a deliberate sign change in a nonzero quarter-flux edge phase, which failed the Gram constraint as it should. These checks corroborate the mathematical claims and the matrix-level interpretation of the new formal material, without being a replacement for the author's version-pinned build.

The journal contribution is appropriately narrow. Its graph and general monopole technique are known; its distinctive work is the orbit-sensitive decomposition, a useful skew operator, the explicit quarter-flux gauge and intertwiner, and the two special-flux factorizations. Publication readiness is now chiefly conditional on the independently replayable frozen release and permanent archival citation. The existing figures are correctly described as connections between sorted discrete spectra, not analytic branches.

1 Principal positive findings

1.1 Ordinary spectrum and its representation structure

The submission defines the graph on six square faces and eight hexagonal faces. Its induced hexagon graph is the cube Q_3 , while the square–hexagon incidence connects (a, σ) to the four $h \in \{\pm 1\}^3$ with $h_a = \sigma$. This is the correct starting incidence description. The decomposition

$$\mathbb{R}^{14} \cong 2A_{1g} \oplus E_g \oplus 2T_{1u} \oplus T_{2g} \oplus A_{2u}$$

is dimensionally complete. In the orthonormal T_{1u} multiplicity basis the Laplacian block is

$$\begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}, \quad \det(xI - L_{T_{1u}}) = x^2 - 9x + 16.$$

The A_{1g} block contributes $x(x-7)$; E_g, T_{2g}, A_{2u} contribute respectively $(x-4)^2, (x-7)^3, (x-9)$. Hence the manuscript’s characteristic polynomial (2) follows without a 14-by-14 determinant. This algebra and the degree, dimension and trace controls are consistent with prior examined versions. It is *inherited established evidence*, not a new execution of the author’s code for Draft 9.

The distinction between the T_{1u} action $K^2 = -4I$ and the A_{1g} action $K^2 = -12I$ is well drawn. Particularly valuable is the explicit warning in Remark 5.2 that K does not annihilate the full eigenvalue-7 space, since that space contains the A_{1g} component.

1.2 The magnetic spectral classes

Let A_q be the Hermitian magnetic adjacency and $\Delta_q = D - A_q$. The 24 oriented triangular plaquettes all carry phase $\exp(i\pi q/12)$. Independently of the chosen gauge, the closed-walk calculation underlying Proposition 6.2 reads

$$\mathrm{tr}(D^3) = 6 \cdot 4^3 + 8 \cdot 6^3 = 2112, \quad \mathrm{tr}(DA_q^2) = 6 \cdot 4^2 + 8 \cdot 6^2 = 384,$$

$$\mathrm{tr}(A_q^3) = 24 \cdot 6 \cos(\pi q/12) = 144 \cos(\pi q/12),$$

and thus

$$\mathrm{tr}(\Delta_q^3) = 3264 - 144 \cos(\pi q/12). \tag{1}$$

Equal spectra imply equal third moments, and the cosine distinguishes the representatives $q = 0, \dots, 12$. The identification $q \sim 24 - q$ follows from complex conjugation. This is a compelling and economical classification *once the spherical flux/gauge classes have been established*.

The claimed quarter-flux factorization follows from the displayed identity

$$B^*B = \frac{9I - C^2}{2} = 4(P_1 + P_2), \quad BB^* = 4I_6.$$

For cube eigenvalues $\mu = 1, -1$, the respective 2×2 blocks are

$$\begin{pmatrix} 4 & -2 \\ -2 & 5 \end{pmatrix}, \quad \begin{pmatrix} 4 & -2 \\ -2 & 7 \end{pmatrix},$$

which yield $(x^2 - 9x + 16)^3$ and $(x-3)^3(x-8)^3$, with two scalar sectors completing the count. This explains the persistent irrational pair at $q = 6$. At $q = 12$, the all-negative edge gauge gives the signless Laplacian $D + A$; the statement is exact and requires no numerical extrapolation. These results remain unchanged from Draft 8; their earlier exact corroboration is not being counted as a new clean-tag replay.

2 The new Lean claims: statement-level assessment

2.1 What is actually formalized

The paper’s Table 2 and Section 8 distinguish ordinary hand proofs, exact symbolic computation, numerical corroboration, and Lean kernel checks. This distinction is essential and largely well executed. Inspection of the public Lean sources at `face-graph-note-v9` identified the following theorem statements and limitations.

Source file	Claimed checked result	What remains outside the file
<code>Theorem31.lean</code>	Explicit 14×14 rational Laplacian L_{fin} with characteristic polynomial $x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3$. Uses an explicit invertible basis and seven small blocks.	Geometric construction and O_h character identification are justified separately in prose.
<code>Theorem51.lean</code>	$K = P_s L P_h - P_h L P_s$ for explicit literals; K acts by the displayed blocks in the symmetry-adapted rational basis, has $K^2 = -4I$ on the three odd planes and $K^2 = -12I$ on the orbit-constant plane.	O_h -isotypic identification (Lemma 3.2); part 5.1(3), singular values between irrational eigenspaces.
<code>QuarterFlux.lean</code>	Characteristic polynomial $(x-9)(x-8)^3(x-3)^4(x^2-9x+16)^3$ for the explicit Gaussian-integer $q = 6$ matrix; an associated Gram identity.	General $U(1)$ face-holonomy classification, and all other flux sectors.
<code>HalfFlux.lean</code>	Characteristic polynomial $(x-8)^3(x-5)^3(x-4)^2(x-3)^4(x^2-13x+24)$ for the explicit signless Laplacian.	Proof that all conceptual interpretations of the graph and flux follow from the formalized literals.

This is a meaningful development beyond a floating-point comparison. For example, the visible conclusion of `Theorem31.lean` is `theorem charpoly_Lfin`, and the visible conclusions of the flux files are `charpoly_D6` and `charpoly_Sfin`. The source shows explicit use of characteristic-polynomial invariance under change of basis, not merely a numerical eigenvalue listing. The quarter-flux proof is stated over $\mathbb{Z}[i]$, with a fraction-field argument for invertibility, consistent with the manuscript’s claim of exact Gaussian-integer input.

2.2 The crucial bridge between a proved literal and the scientific statement

The weakest remaining *formal coverage* link is the passage from a hand-entered matrix to the physical face graph. A kernel can prove `charpoly(Lfin) = p(x)` while a mistakenly entered L_{fin} would still yield a formally valid statement about the wrong object. Draft 9 says its matrices are the same as the manuscript’s equations and the verification scripts. The matrices visible in the public files use the declared ordering (six squares first, eight cube sign-vectors second), and targeted independent reconstruction produces the same expected incidence pattern and polynomial. However, no new Lean theorem formalizes the coordinate-based graph constructor and proves its equality to the literal matrix. This is an explicit *coverage boundary*, not evidence that a matrix was entered incorrectly.

An economical optional strengthening is a small trusted bridge: define graph adjacency by the rules $h_a = \sigma$ and Hamming distance one, prove the literal matrix equals the resulting Laplacian, and document the theorem name. Similarly the explicit orientation and gauge-to-literal- Δ_6 correspondence could be formalized, but that is not required to make the present hand proofs sound. It would substantially improve the meaning of “machine checked” for a nonspecialist reader.

2.3 Axiom and release trust status

Section 8 reports pinned Lean/Mathlib v4.34.1, no `sorry` or `native_decide`, and `#print axioms` outputs restricted to `propext`, `Classical.choice`, and `Quot.sound`. The public code contains explicit theorem declarations and axiom-print commands consistent with that description, but the output of an independent `lake build` was not obtained. In particular, *inspection of source is not the same event as reproducing a kernel-checked build*. The reported axiom list is provisionally credible, not independently certified here. A clean checkout and build log must be retained with the definitive archive.

3 New exact arithmetic and negative control

For independent corroboration of the mathematical object to which the formal sources refer, construct A on square nodes (a, σ) and hexagons $h \in \{\pm 1\}^3$ by $A_{(a, \sigma), h} = 1$ iff $h_a = \sigma$, and $A_{h, h'} = 1$ iff the Hamming distance is one. Thus $L = D - A$ and $S = D + A$. Direct exact characteristic-polynomial calculation gives

$$\begin{aligned}\chi_L(x) &= x(x-9)(x-7)^4(x-4)^2(x^2-9x+16)^3, \\ \chi_S(x) &= (x-8)^3(x-5)^3(x-4)^2(x-3)^4(x^2-13x+24).\end{aligned}$$

From the printed quarter-flux gauge, the 6-by-8 block B has entries in $\{0, \pm 1, \pm i\}$ and exactly satisfies $B^*B = (9I - C^2)/2$ and $BB^* = 4I_6$. Assembling the 14×14 Hermitian matrix in equation (8) yields its full polynomial by exact algebra; a one-entry sign perturbation of B breaks the Gram equation. The perturbation is a test-sensitivity control, not a physical counterexample.

For K we independently calculated $K^T = -K$, $Ks_a = h_a$, $Kh_a = -4s_a$ for all three axes, and $K^2 = -12I$ on orbit constants. These are the same transparent basis-level facts addressed by the new Lean file, though the singular-value conclusion of Theorem 5.1(3) is established by the manuscript's orthonormal-plane argument rather than this new formal source.

The 24 oriented triangular faces of the actual embedded graph give $\partial_1\partial_2 = 0$, $\text{rank } \partial_1 = 13$, $\text{rank } \partial_2 = 23$, and sum of their outward boundaries equal to zero. Their image is therefore the entire cycle space. This checks the new text of Proposition 6.1 without claiming that Proposition 6.1 was newly formalized in Lean. A total of 25 targeted algebraic assertions passed; that figure counts automated checks, not 25 independent theorems or reviews.

4 Topological scope and attempted falsification

Draft 9 has incorporated the most important control from the previous review directly into Proposition 6.1 (page 6). Its proof now states the chain-complex incidence ranks and the equality of the image of ∂_2 with the graph's cycle space, not just a counting heuristic. The mathematical implication is

$$\text{rank}(\partial_2) = 23 = 36 - \text{rank}(\partial_1) \implies \text{im}(\partial_2) = \ker(\partial_1).$$

Two edge-phase assignments with identical face holonomies have a quotient connection with trivial holonomy on every face. Because every closed loop lies in the face-boundary span on this spherical embedding, the quotient has trivial holonomy on all cycles and is pure vertex gauge. The conclusion applies to the stated cellularly embedded sphere, not to arbitrary embeddings of an abstract graph.

The manuscript also gives an appropriate torus negative control. In a periodic four-by-four square grid, the horizontal edge phase $\exp(i\theta/4)$ yields identity holonomy around every elementary square but $\exp(i\theta)$ around a noncontractible winding loop. Hence flat local plaquette holonomy alone does not classify toroidal connections. The counterexample does not refute Proposition 6.1, because its hypothesis is S^2 . The current formulation is materially stronger and better communicated than Draft 8; no topology-related correction remains open.

5 Novelty, provenance and reproducibility

5.1 Priority now appropriately delimited

The expanded introductory paragraph on page 2 now distinguishes the existing graph, its automorphism group and adjacency polynomial (MathWorld) from the known monopole-on-spherical-graphs method (Avishai and Luck, 2008). The present contribution is the graph-specific O_h Laplacian decomposition, K , the two special-flux factorizations, quarter-flux gauge and explicit intertwiner. The targeted background search for a magnetic spectrum of this exact graph did not identify a publication of the same factorization; this is bounded non-detection, not proof of absolute priority. Thus the previous scholarly correction C2 is **closed** by Draft 9's revised language.

5.2 Immutable tagged-release execution is still the decisive external check

The submitted PDF names `face-graph-note-v9` and documents the Python runners, numerical tolerances, integer/gauge certificate and pinned Lean 4/Mathlib environment. The public release page identifies that tag, and the linked source files can be inspected as text. The full tagged repository archive could not be transferred into this execution environment; Lean/Lake are not installed here. Accordingly no reviewer-executed `lake build`, original Python one-command runner, proof-axiom log, or Zenodo-deposit hash comparison is claimed.

For acceptance-level reproduction, one should check out the exact tagged commit in an isolated environment, record its checksum, install the pinned Python and Lean dependencies, execute `python verification/run_face_graph_checks.py` without `-O`, then execute `lake exe cache get` && `lake build` from the Lean project directory. Retain the independent exit codes, visible axiom outputs, regenerated certificate, flux-sweep CSV, figure and a hash manifest. The article's permanent archive identifier should point to precisely this release. A successful author-side replay is welcome evidence but is not equivalent to an independent replay.

5.3 What changed from Draft 8

The characteristic polynomials, Gaussian-integer gauge and intertwiner, magnetic spectrum-class count, and Fedorov table are inherited. The changes relevant to the verdict are: (i) an expanded and better-bounded novelty paragraph; (ii) promoted boundary-rank calculation and toroidal counterexample; (iii) a new, scoped Lean proof ledger; (iv) updated acknowledgement and reproducibility detail. There is no justification for reopening established errors or representing inherited results as newly discovered. The fresh checks in this review target the mathematical identities newly claimed as formally verified and the graph/matrix correspondence.

6 Fourteen-metric scientific assessment

Values are diagnostic judgments on a ten-point scale, not probabilities. This specialist note is evaluated under the theoretical mathematical-physics profile (core category weights: correctness 40%, completeness 25%, clarity 20%, significance 15%). An aggregate number is not used. The within-category emphasis follows the defined review rubric.

ID	Criterion	Score	Manuscript-specific justification
M1	Theorem and claim precision	9.0	Main theorems, literal-matrix scope, and non-formalized extensions are individually identified.
M2	Proof and support completeness	9.0	Hand proofs plus meaningful Lean source; claimed kernel pass not independently replayed.

ID	Criterion	Score	Manuscript-specific justification
M3	Calculation and analysis accuracy	9.0	Twenty-five fresh symbolic assertions pass, including a deliberately perturbed-phase control.
M4	Logical consistency	9.0	Exact spherical boundary-span and toroidal negative case distinguish the domains.
M5	Definition coverage	9.0	Basis, flux gauge, matrix ordering, proof scope and numerical branches are defined.
M6	Assumption transparency	9.5	Clear separation of hand proof, exact computation, Lean literals and not-yet-formalized steps.
M7	Scope specification	9.0	Narrow contribution and formal coverage are well delimited; tagged replay outstanding.
M8	Notation consistency	8.5	Matrix conventions are specified, with some visually dense formulas.
M9	Proof accessibility	9.0	Small orbit blocks and structured Lean proof sources support independent scrutiny.
M10	Organization/navigation	9.0	Expanded prior-art text and Table 2 make proof status easy to locate.
M11	Originality	7.5 (prov.)	Specific magnetic factorization not detected in bounded search; priority cannot be certified exhaustively.
M12	Generality	6.5	Results concern one graph and exact charge 0, 6, 12, despite broadly useful methods.
M13	Literature/attribution	9.0	Revised Introduction distinguishes catalogued graph from prior magnetic methodology.
M14	Potential impact	7.0	Valuable specialist example; no broader physical law or cell-selection theorem claimed.

Category interpretation: mathematical correctness and supporting proofs are strong; completeness is high but external computational corroboration remains conditional; clarity is strong with minor presentational demands; significance is that of a potentially original *specialist example*, not a demonstrated general principle.

Specialized checks (not added to an aggregate): M15 Gauge/invariant structure 9.0/10 — the spherical boundary rank and negative control support the stated hypothesis; M18 Algebraic structure 9.0/10 — the exact orbit blocks, magnetic matrices and declared formal scope are explicit. M16 (claimed breakthrough) and M17 (interdisciplinary synthesis) are not activated by this deliberately mathematical note.

7 Remaining closure requirement and dependencies

R1 — Independent reproduction of the definitive release (required). Supply the immutable deposit reference matching tag `face-graph-note-v9`; independently execute the entire four-script Python runner and pinned Lean/Lake build; save its actual output and axiom prints with commit identification and generated-file checksums. A failure limited to unavailable packages or transient environment configuration is not a mathematical counterexample but requires a corrected reproducibility recipe. A reproducible mismatch in the literal matrix, characteristic polynomial, or specified holonomies would reopen the relevant theorem and could reverse the recommendation.

R2 — Formal statement bridge (optional). Clarify in one sentence, or optionally formalize, that Lean’s literal L_{fin} and D_6 are identified with the geometric graph and the oriented gauge of Sections 2 and 6. This distinction is already mostly made; no new Lean theorem is required merely to retain the current hand proofs. Avoid saying the entire representation-theoretic or gauge-uniqueness results are machine-checked, since the supplied source explicitly leaves them outside its formal scope.

Previously requested modifications closed. The prior-art separation is now explicit. The Proposition 6.1 rank proof and toroidal negative control are incorporated. The inherited proof of

the 13 distinct charge spectra is still independent of Lean and rests on the third-moment argument following spherical gauge classification. Failing to compile `Theorem51.lean` would affect formal corroboration of the K blocks, not automatically its short printed hand derivation. Failing to match the $q = 6$ literal gauge matrix would be materially more serious, since it calls into question the connection between the magnetic theorem and its intended plaquette holonomies.

8 Recommendations for authors and editor

Authors. Complete R1 before final acceptance and cite the exact Zenodo archive. Retain Table 2’s current proof-coverage distinctions. If feasible, include a one-page mapping from Lean theorem names to equations (2), (5), (6), (9), and the specified K identities; a single reproducible terminal log is more valuable than more parallel numerical confirmations. Preserve the graph-specific, non-universal novelty language and the fact that the Figure 1 segments are not analytic eigenvalue branches.

Editor. The manuscript deserves a favorable *conditional minor-revision* assessment as specialist finite mathematics. The known graph and general magnetic method are not offered as discoveries; the quarter-flux construction and exact graph-specific formulas are the distinctive work. Prioritized release validation is the only outstanding clearance gate. The visible Lean artifacts strengthen evidentiary support but do not justify marking the entire paper “formally verified” until the tagged build has been independently replayed and the claimed axioms reviewed.

What would reverse this recommendation? A verified disagreement between the matrices specified by the graph and gauge of the manuscript and the literal matrices asserted in the Lean theorems, or a valid spherical counterexample to Proposition 6.1, would elevate the decision to major revision or rejection depending on the affected claims. A persistent, unexplained failure of the tagged code or Lean build would prevent release clearance. By contrast, discovery of an identical earlier factorization would affect originality and journal fit more than correctness. Independent success of the complete tagged replay and a fixed archive citation would remove the outstanding evidentiary condition.

9 Final judgment

Draft 9 improves the scientific paper in precisely the areas singled out by the preceding round. Its central finite-matrix mathematics remains consistent with fresh exact checks, its topological assumptions are carefully stated, and its new Lean sources provide substantive formal proof artifacts with appropriately disclosed boundaries. **Recommendation: conditional minor revision.** The remaining prerequisite is a clean, independently recorded replay of the definitive Python and Lean release, not correction of any currently confirmed major mathematical error.

References inspected for this review

- [R1] E. W. Weisstein, *Tetrakis Hexahedral Graph*, Wolfram MathWorld, <https://mathworld.wolfram.com/TetrakisHexahedralGraph.html>. Catalogued graph, automorphism order and ordinary adjacency polynomial.
- [R2] Y. Avishai and J. M. Luck, *Tight-binding electronic spectra on graphs with spherical topology I: The effect of a magnetic charge*, J. Stat. Mech. (2008) P06007, DOI [10.1088/1742-5468/2008/06/P06007](https://doi.org/10.1088/1742-5468/2008/06/P06007). Closely related magnetic-flux setting and examined examples.